In [2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [5]:
```
df=pd.read_csv('loan.csv')
df.head()
```

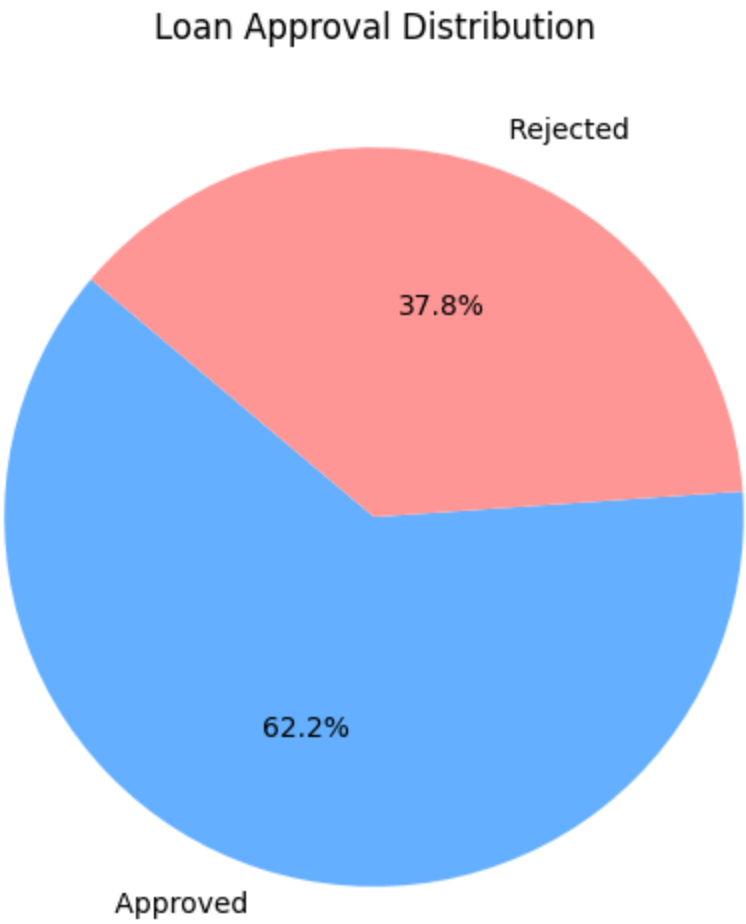
Out[5]:

	loan_id	no_of_dependents	education	self_employed	income_annum	loan_amount	loa
0	1	2	Graduate	No	9600000	29900000	
1	2	0	Not Graduate	Yes	4100000	12200000	
2	3	3	Graduate	No	9100000	29700000	
3	4	3	Graduate	No	8200000	30700000	
4	5	5	Not Graduate	Yes	9800000	24200000	

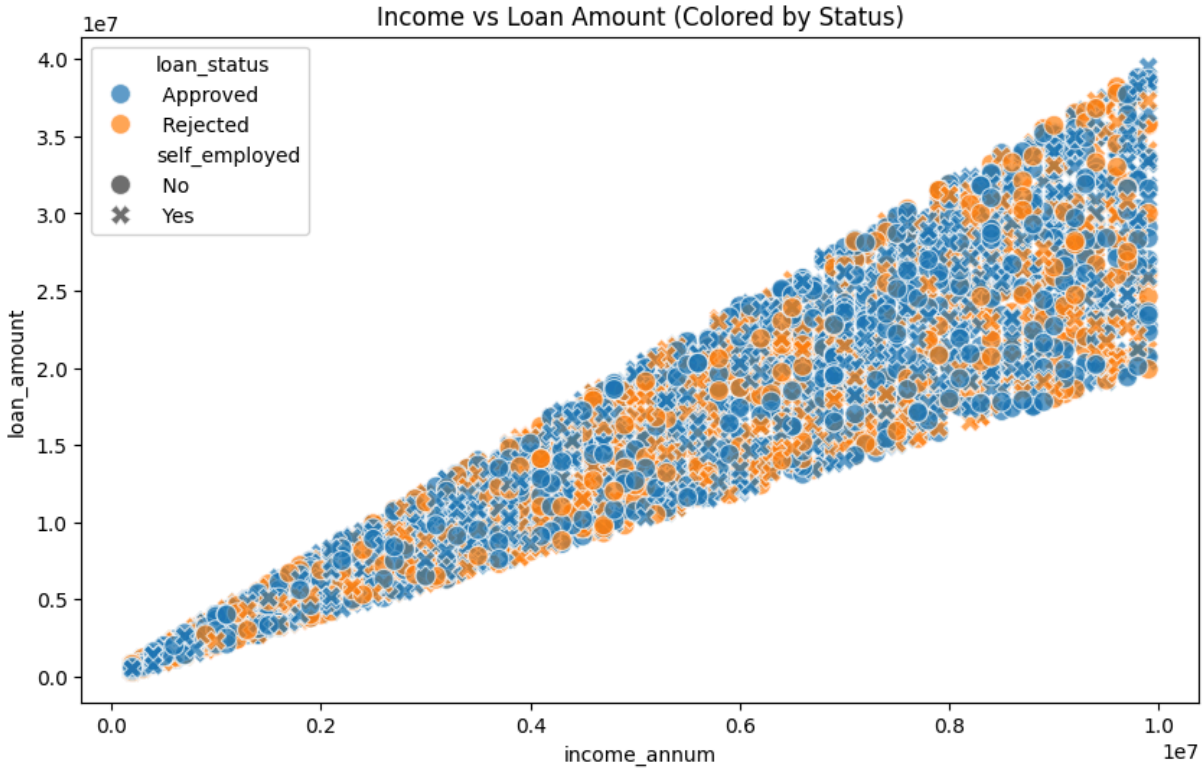
In [11]:
```
df_new=df.drop('loan_id',axis=1)
print(df_new.head())
```

```
   no_of_dependents      education self_employed  income_annum  loan_amount  \
0                 2       Graduate            No       9600000     29900000
1                 0   Not Graduate           Yes       4100000     12200000
2                 3       Graduate            No       9100000     29700000
3                 3       Graduate            No       8200000     30700000
4                 5   Not Graduate           Yes       9800000     24200000

   loan_term  cibil_score  residential_assets_value  commercial_assets_value  \
0         12          778                   2400000                 17600000
1          8          417                   2700000                  2200000
2         20          506                   7100000                  4500000
3          8          467                  18200000                  3300000
4         20          382                  12400000                  8200000

   luxury_assets_value  bank_asset_value loan_status
0             22700000           8000000    Approved
1              8800000           3300000    Rejected
2             33300000          12800000    Rejected
3             23300000           7900000    Rejected
4             29400000           5000000    Rejected
```

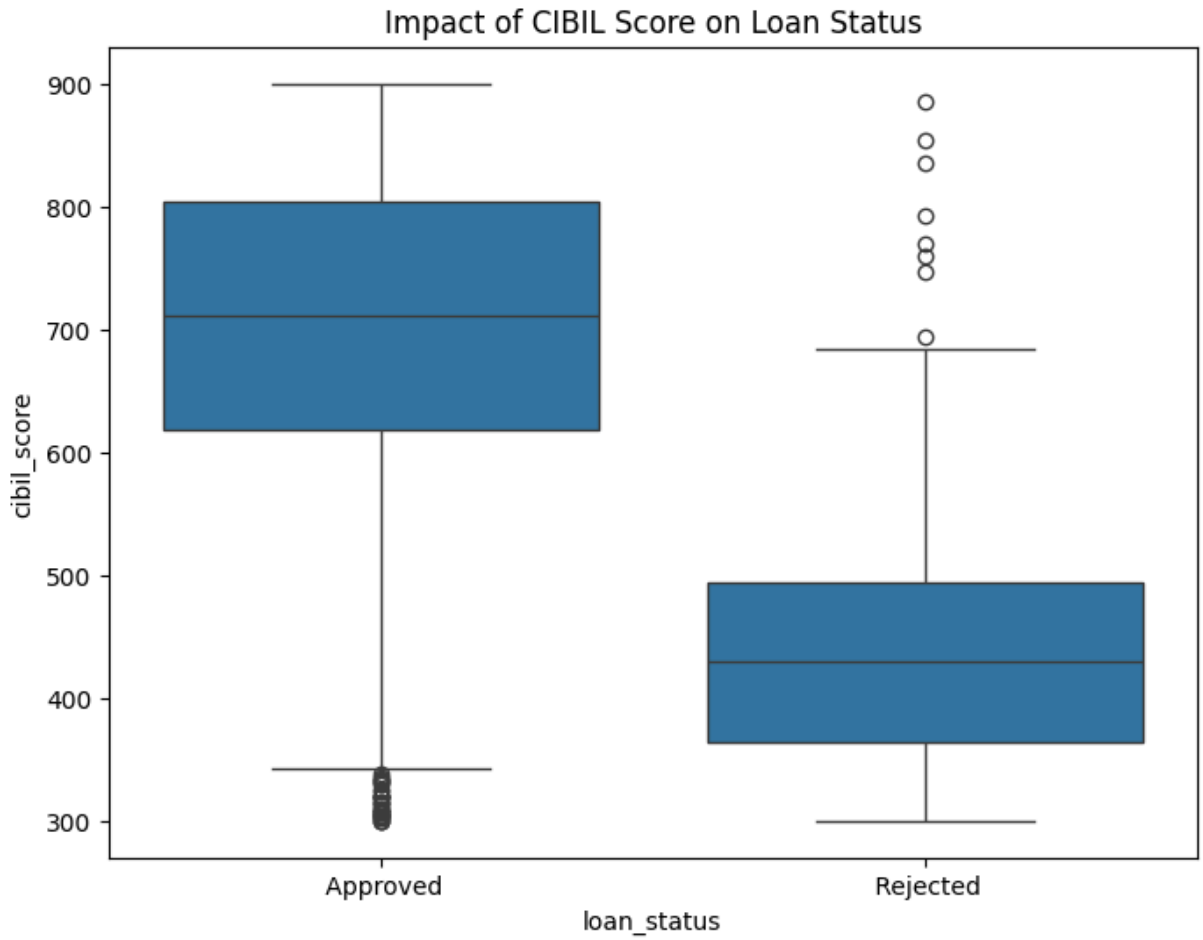
In [14]:
```
df.columns = df.columns.str.strip()
plt.figure(figsize=(6, 6))
# Count the values in 'loan_status'
status_counts = df['loan_status'].value_counts()
plt.pie(status_counts, labels=status_counts.index, autopct='%1.1f%%', startangle=14
plt.title('Loan Approval Distribution')
plt.show()
```



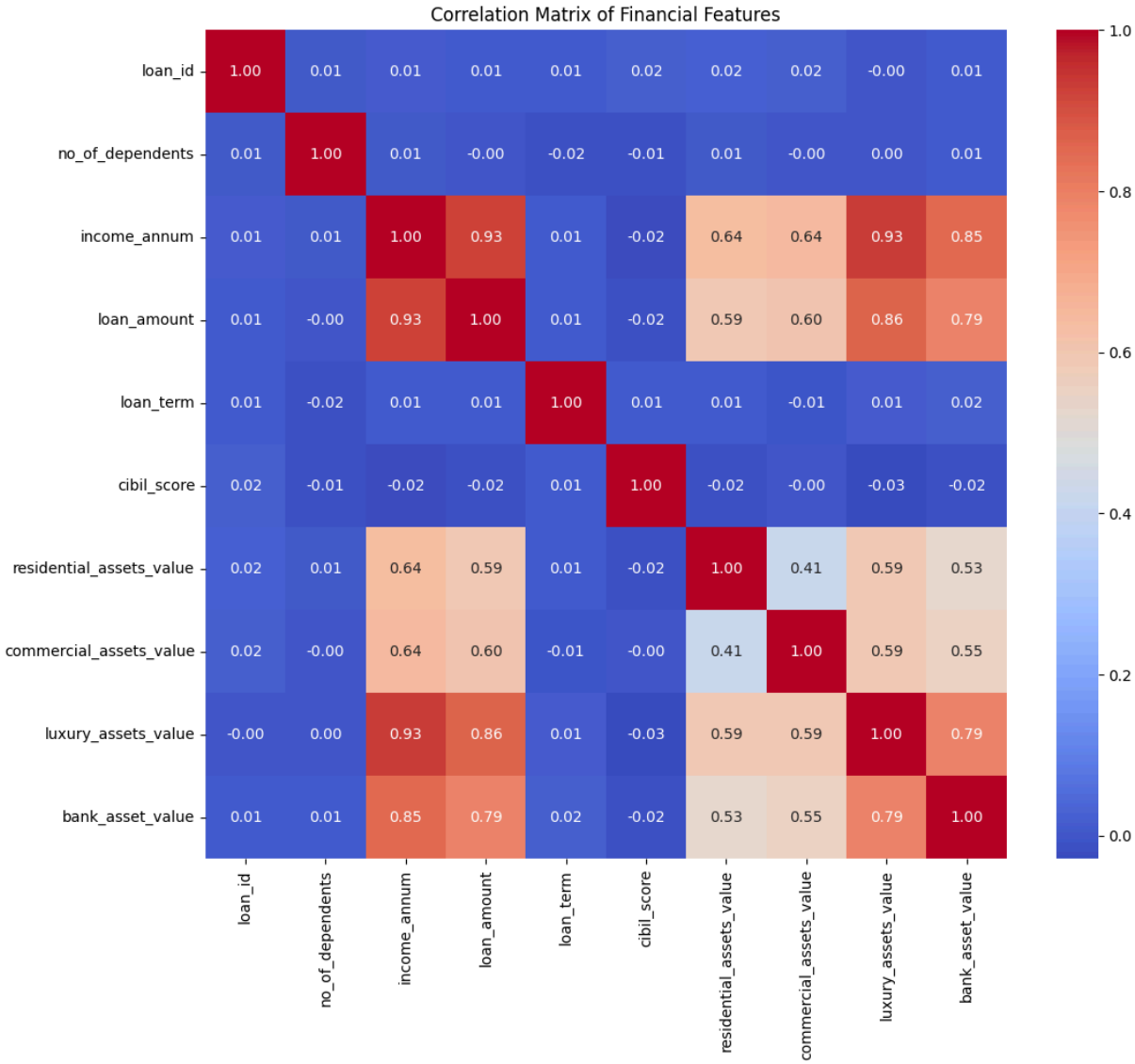
In [22]:
```
plt.figure(figsize=(10, 6))
sns.scatterplot(
    data=df,
    x='income_annum',
    y='loan_amount',
    hue='loan_status',  # Colors points by Approved/Rejected
    style='self_employed', # Different shapes for Self Employed or not
    s=100,
    alpha=0.7
)
plt.title('Income vs Loan Amount (Colored by Status)')
plt.show()
```



In [27]:
```
plt.figure(figsize=(8, 6))
sns.boxplot(data=df, x='loan_status', y='cibil_score')
plt.title('Impact of CIBIL Score on Loan Status')
plt.show()
```

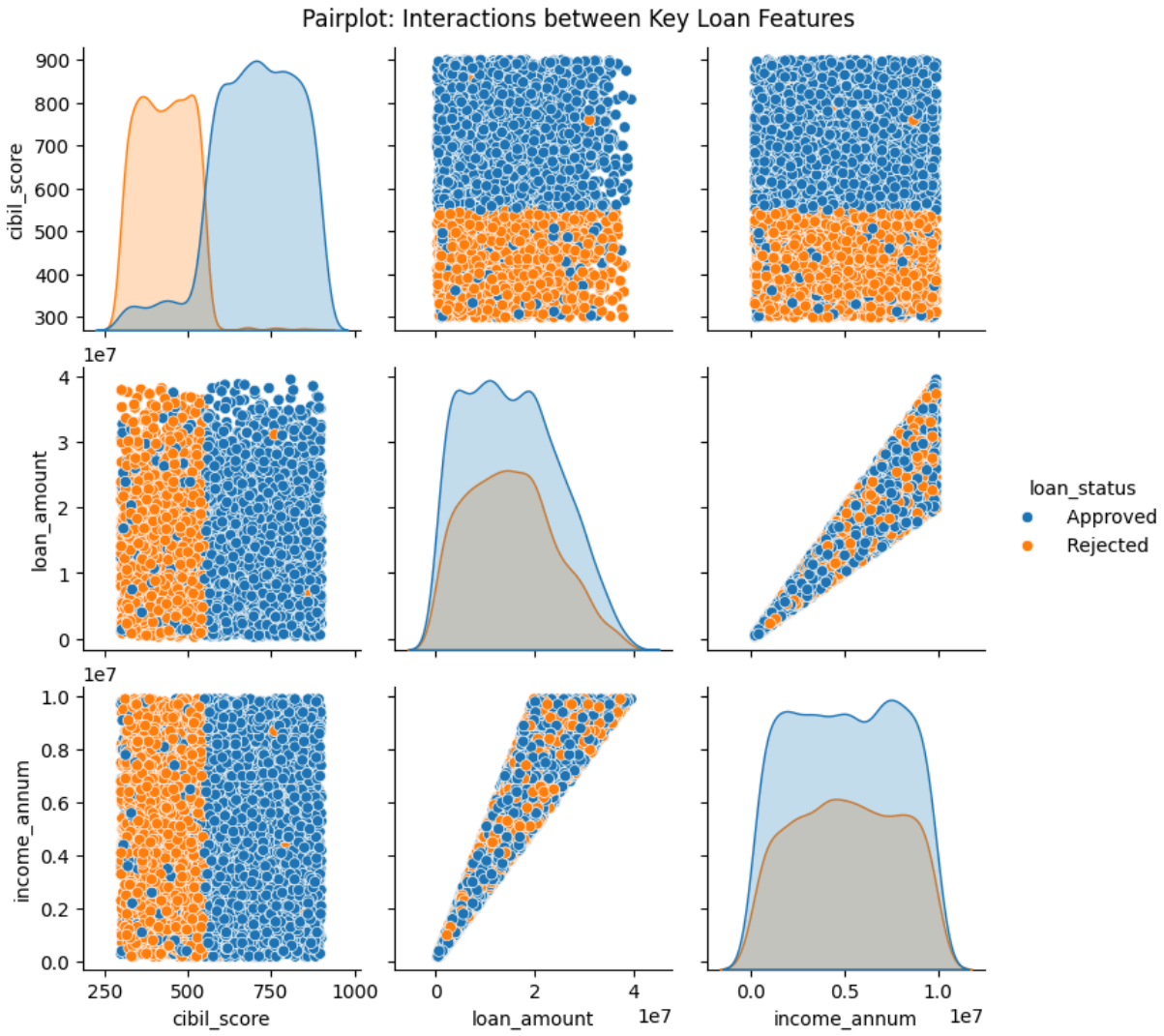


In [29]:
```
plt.figure(figsize=(12, 10))

# We drop non-numeric columns for the correlation matrix
numeric_df = df.select_dtypes(include=['float64', 'int64'])
corr_matrix = numeric_df.corr()

sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt=".2f")
plt.title('Correlation Matrix of Financial Features')
plt.show()
```



In [35]:
```
cols_to_plot = ['cibil_score', 'loan_amount', 'income_annum', 'loan_status']
plt.figure(figsize=(10,8))
sns.pairplot(df[cols_to_plot],hue='loan_status',diag_kind='kde')
plt.suptitle('Pairplot: Interactions between Key Loan Features', y=1.02)
plt.show()
```

```
<Figure size 1000x800 with 0 Axes>
```


Pairplot: Interactions between Key Loan Features
cibil_score
loan_amount
income_annum
1e7
loan_status
Approved
Rejected